

Source Documentation

Global Game Jam 2026: Hardware & Software

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1. Introduction

This documentation serves as the collection of information between the hardware and software of this project.

1.1. Important Dates and Timeline

The entire game was made on a single weekend, from January 30th to February 1st. The hardware construction began 2 weeks earlier. To stay within the nature of the competition, the game can be played, and is presented in a more typical “.exe” format and in a web browser on The Game Jam Website. The Game was finished on Feb. 1st, 2026. We continued to work and develop hardware into the following week.

2. Technology Used

2.1. Hardware

2.1.1. Raspberry Pi Pico 2 WH

We went with the Raspberry Pi Pico 2W as the “brain” of our demonstration. Inspired by the Pico GB project, we wanted to take the core reverse engineering provided by the project and put our own game and modifications onto the project.

2.1.2. Standard Push Buttons

As a cost saving measure, we got a large pack of standard push buttons as all inputs for the hardware.

2.1.3. 22 Gauge Wire

We used this wire because it was available and thicker than the original gauge wire used by Pico GB, any gauge thicker than 24AWG will do.

2.1.4. 2.2" LCD Screen

While not the original model used by Pico GB, we matched their specs exactly, 2.2inch diagonally with 220p x 176p resolution. Any screen with these specs will do. It is important that they have an SD card slot for game reading.

2.1.5. Solder Boards

To save on weight and thickness we used some donated solderboard with a matching connector layout as a breadboard. This allowed us to match our prototyping, cut cost, and save time when building the final model. Any breadboard, or (preferably) solderboard, will do.

2.1.6. Speakers

We went with an 8ohm resistive speaker for sound. Any wattage from 1-2 Watts will do. It should be noted that higher the wattage the louder the sound.

2.1.7. Audio Interface and Amplifier

We went with a standard I2C sound board for all audio interfacing.

2.1.8. TI-84 Plus CE python edition

As an emergency backup, we had this calculator on hand in case our initial hardware plans failed.

2.1.9. Total Hardware cost

Item Cost	
Raspberry Pi Pico 2 WH (2 pack)	\$11.98
Standard Push Buttons (Large bundle)	\$9.99
22 Gauge Wire	\$5.99
2.2" LCD Screen (2 pack)	\$26.50
Solder Boards (3 pack)	\$4.99
Speaker (2 pack)	\$11.98
Audio Interface (3 pack)	\$11.99
Total Cost	\$83.42

Table 1: Total Cost Breakdown

2.2. Software

2.2.1. Gb Studio

This was our go-to game engine for designing, testing, and compiling/presenting our game to the game jam community.

Link: <https://www.gbstudio.dev>

2.2.2. Aseprite

This was the general pixel art program used to create the tilemaps for the map. While this software has a free version, the team already had access to a steam liscense.

Link: <https://www.aseprite.org/>

2.2.3. TileD

This software was used in alongside Aseprite to create the tilemaps. This software is completly free for all major platforms.

Link: <https://www.mapeditor.org/>

2.2.4. Krita

We used this open-source art software to make some of the character sprites and actors.

Link: <https://krita.org/en/>

2.2.5. Solidworks/FreeCAD

For all 3D models for the Gameboy cases, we used a variety of solidworks/FREECAD to design, view, and slice in order for us to 3D print what we needed. All models are included in section 3.

Link: <https://www.solidworks.com>

Link: <https://www.freecad.org>

2.2.6. Cura Slicer

For slicing models created for the gameboy cases, we decided to go with cura slicer in order to print. For printers the following were available:

- Creality Ender V3 SE
- Creality Ender V2 Standard

2.2.7. Git

As seen with this repository, git was the version control system the team agreed on. We then uploaded the repository to github.

Link: <https://git-scm.com>

Link: <https://github.com/JEvan234/Global-Game-Jam-2026>

2.2.8. Total Software Cost

All of the software used for this project was free and open source software (FOSS). Outside of solidworks, which has

alternatives like FREECAD, none of the software has a purchase or liscensing fee.

3. Schematics and Models

3.1. Printed Shell

3.2. Hardware Internals

3.3. Git Workflow

4. Prior to Day One

4.1. Buying the Hardware

We bought our hardware from a variety of storefronts ranging from our local Micro Center to Amazon online.

4.2. Checking Competition Legality

According to the Game Jam Website, link at bottom, All alternative-hardware titles were competiton legal and welcome.

Link: <https://globalgamejam.org/jam-sites/2026/kennesaw-state-university>

4.3. Preperations

Leading into day one of the Game Jam:

- Initialized the repository and documentation
- Acquired materials
- Downloaded software
- Assembled team and established goals for the jam

5. The Game Jam

5.1. Theme

The Game jam Theme this year is **Mask**

5.2. Our Game

Veil of Dusk and Dawn is a asymmetrical Co-Op multiplayer title meant to be ran on any hardware capable of gameboy emulation. You play as 1 man split across time as you attempt to return a magical mask back to it's rightful place, reuniting the timeline. One player is in the "Dusk" form, meaning he cannot see with the mask on, but can interact with the world, while in this form the player should be careful to not set off the alarms at every exhibit. To help in navigate, a second player named "Dawn" can see the world, but cannot interact with the world as in his timeline the mask is already returned. Dawn can use his vision to guide the Dusk player to returning the mask.

5.2.1. Art Style

The art and sprites for the game was inspired from pokemon top-down characters. We used a man with a masked and unmasked version, wearing a fedora to add character. Background and sprites follows the theme and story of the game, which takes place in a museum. All sprites followed a 16px by 16px tile format, while the map tilesheets used 8px square tiles.

5.2.2. Music Style

The music for Veil of Dusk and Dawn came from a leitmotif that we thought would fit the theme of the game being

masks. the night menu track was first and the first expression of said leitmotif, and then the day is a more laid back version, with both in 8/8. For the gameplay theme of both, to add more intensity we switched to 7/8 time. For the winning music to show completion and finality we pushed it back further to 6/8 time. All music is available on youtube:

Link: <https://www.youtube.com/playlist?list=PL-Wx1DP4eIDqWribw59Uyw-Q9BC5yOx4l>

5.2.3. Screenshots

Attached are a few game screenshots, order is random and not a reflection of gameplay:

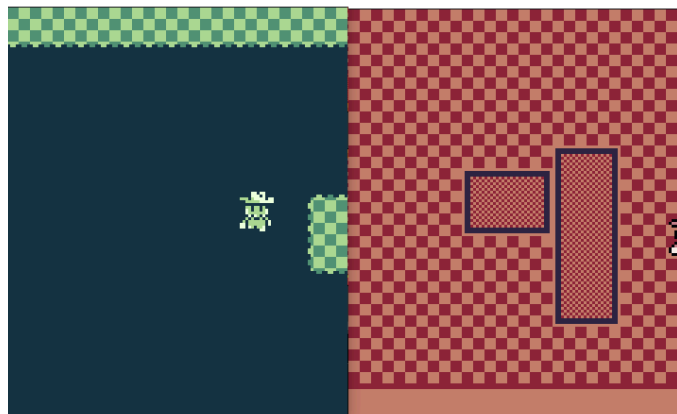


Figure 1: Game Screenshot, Night left, Day right

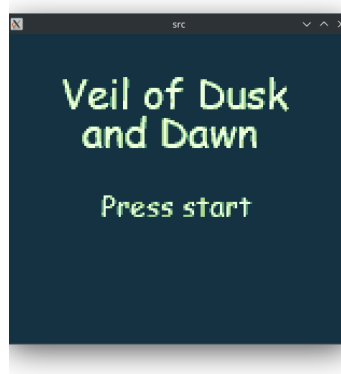


Figure 2: Title Screen

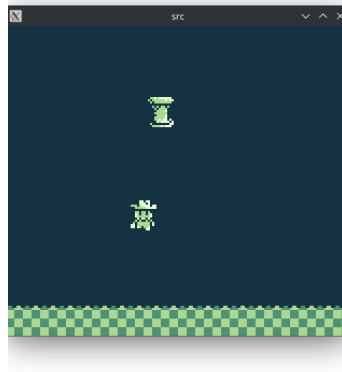


Figure 3: Mask Room, with Pedestal

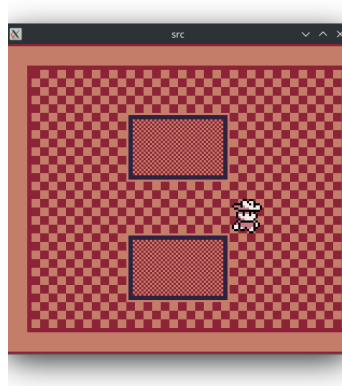


Figure 4: Day Mode

5.3. Jam Timeline

This is our total timeline of events **Within the Game Jam itself**. Refer to the introduction for the timeline for hardware and early development. Attached with each days summary is a list of key events for the day. Days end at 11:59 PM for the date.

5.3.1. Day 1

This day was mostly prep and brainstorming. Key events include:

- Sharing the Repo with the team
- Learning the theme and initial brainstorming

- Agreeing on the game premise
- Dividing initial work
- Beginning final assembly of hardware (All parts were available)

5.3.2. Day 2

This day the real work began, with initial art and music made, and hardware development on the case began.

- Music made
- Art made
- Tilemaps made
- Case Designed
- First case 3D print (Failed)
- Logic was implemented, both versions (Dawn and Dusk) were initially playable

5.3.3. Day 3

This was the final day, with the submission made at 5PM. The final case was printed in the early hours of the morning, and the game was polished and implemented on a TI-84 Calculator as a backup since the first gameboy design did not run for presentation time.

- Finished the game
- Reimplemented night-time tilemap for better visibility
- Presented
- Implemented the game onto the calculator
- Assembled V1.0 of the Gameboy (Case and all), it did not run

5.3.4. Presentation Time

6. Results and Reflection

6.1. Failures

6.2. Successes

6.3. Event Failures

6.4. Overall Takeaway

7. Next Steps

Included below are the next steps we are taking to polish the work we completed for the game jam. There are no deadlines, nor promise of completion. All that is included here is a goal for us to continue developing this project into what we initially pictured before the jam.

7.1. Gameboy Version 2

We plan to continue working on the gameboy and primarily redo the outer 3D printed shell to be thinner (using a solderboard instead of breadboard) and utilize a cleaner design.

7.2. More Games

If we have the time a few people might develop more games for our handheld on GBstudio. It is also possible to find other ROMs for it and install them via the SD card.

8. Credits

A special thanks to:

- Mert Acar (Software, Game Logic)

- Oluwajomiloju Adediran (Software, Game Logic)
- Jacob Evans (Hardware, Software, Documentation)
- Raam Patel (Hardware, Documentation)
- Tri Pham (Software, Art, Music)

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